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Evolving Towards Artificial-Intelligence-Driven Sixth-Generation Mobile Networks: An End-to-End Framework, Key Technologies, and Opportunities

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Abstract: The incorporation of artificial intelligence (AI) into sixth-generation (6G) mobile networks is expected to revolutionize communication systems, transforming them into intelligent platforms that provide seamless connectivity and intelligent services. This paper explores the evolution of 6G architectures, as well as the enabling technologies required to integrate AI across the cloud, core network (CN), radio access network (RAN), and terminals. It begins by examining the necessity of embedding AI into 6G networks, making it a native capability. The analysis then outlines potential evolutionary paths for the RAN architecture and proposes an end-to-end AI-driven framework. Additionally, key technologies such as cross-domain AI collaboration, native computing, and native security mechanisms are discussed. The study identifies potential use cases, including embodied intelligence, wearable devices, and generative AI, which offer valuable insights into fostering collaboration within the AI-driven ecosystem and highlight new revenue model opportunities and challenges. The paper concludes with a forward-looking perspective on the convergence of AI and 6G technology.

Keywords: 6G; AI-native RAN; end-to-end AI; cross-domain

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1. Introduction

The evolution of communication systems is primarily influenced by service requirements. As shown in Figure 1, the International Telecommunication Union Radiocommunication Sector (ITU-R) report identifies the integration of communication and artificial intelligence (AI) as one of the new use cases for sixth-generation (6G) mobile networks [1,2]. Future 6G mobile networks will pursue higher data speeds at a gigabit-per-second magnitude [3]. To support the vision of 6G networks, AI will become a key enabler for various 6G applications and scenarios. As intelligent applications evolve, next-generation networks are anticipated to thoroughly integrate AI, information technology (IT), and communication technology (CT), which are referred to as AICT in this paper. This transition is converting the communication system from traditional data pipelines into intelligent service platforms [4,5]. In the era of fifth-generation mobile networks (5G), the integration of IT and CT has already been explored through concepts such as cloud–network convergence and service-based architectures [6,7]. Building upon this foundation, AICT is introducing AI technologies to further enhance the intelligence of communication systems.

AICT introduces two key concepts: AI for Network (AI4Net) and Network for AI (Net4AI). AI4Net highlights the use of AI technologies to optimize network performance,

which has been extensively studied in both academic and industrial contexts. The 3GPP Radio Access Network (RAN) and Service & System Aspect (SA) working groups (WGs) have investigated AI use cases, attributes, and lifecycle management for the RAN, the core network, and network management systems [8–12]. Diverse AI-based approaches have been proposed to enhance use cases such as mobility management, load balancing, energy saving, beam management, and interference management [13–17]. For instance, in [13], AI automates and refines cell reselection, cell handover, and base station mobility configuration parameters for mobility management, which can reduce network measurement overheads and improve user service experience. In [17], a model-free distributed self-learning interference suppression (SLIM) scheme for autonomous networks is proposed without information exchange between base stations to reduce manual intervention and minimize the impact of inter-cell interference. However, current discussions on AI4Net lack validation within existing networks. The additional overhead introduced by AI and its actual impact on network performance still require further testing and evaluation.

In contrast, Net4AI focuses on improving network capabilities to support diverse AI applications. In this paradigm, the network not only is the pipeline for traditional data transmission, but also provides the computing resources, data, and models necessary for AI processing. By transforming communication networks into intelligent service platforms, the communication service providers can develop AI-driven applications on top of this infrastructure. While the AI4Net model merely applies AI technologies to optimize network performance, Net4AI can also be exposed as a service to third parties, providing models, computing power, or data analytics to support applications such as autonomous driving, AI-driven industrial Internet of Things (IoT), and smart cities [18,19]. This approach enables telecom operators to expand their revenue models beyond traditional connectivity services.

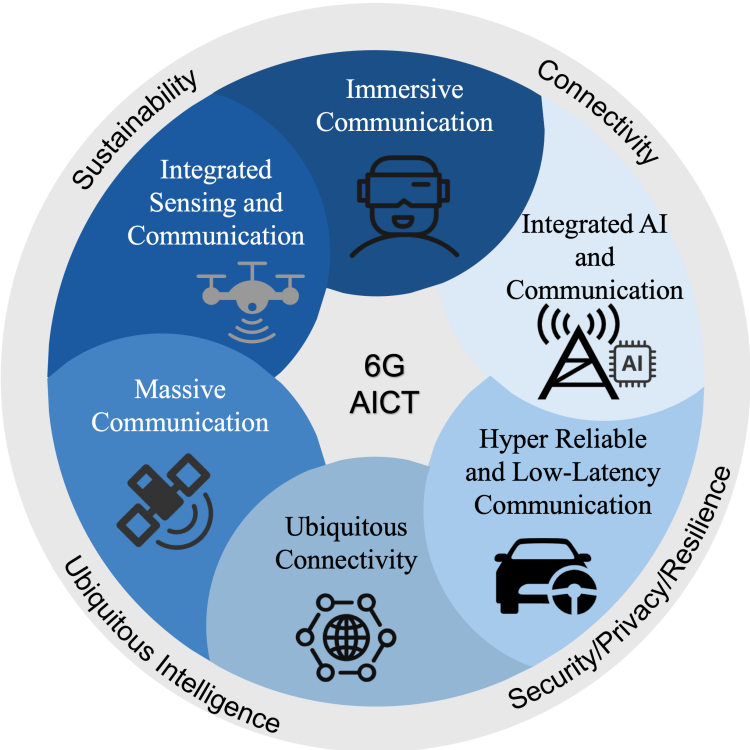


Figure 1. Typical 6G use cases defined by the ITU-R.

In the 5G and 5G-advanced stage, AI and communication systems function more like two independent entities, with AI integrated as an add-on feature to the existing communication networks. Under this approach, AI algorithms must be separately designed

and deployed for each application scenario, potentially leading to redundant network functions and inefficient resource utilization. Additionally, this model relies on centrally deployed AI models, making it difficult to support real-time AI applications with stringent latency requirements, such as intelligent control and autonomous driving. Therefore, AI should be natively embedded into the 6G network from the initial system design. This requires a holistic approach across infrastructure, network architecture, and interface design to enable 6G technology to support both communication and AI-driven services. AI deployment is expected to evolve into a hybrid model combining centralized and distributed intelligence. By leveraging the network's connectivity capabilities, AI functions at different locations within the network can be efficiently orchestrated, providing users with ubiquitous AI services.

However, the path to AICT in 6G networks is fraught with challenges. Dynamic resource allocation across the cloud, the edge, and devices requires advanced scheduling algorithms to balance diverse and conflicting requirements. The lack of standardized protocols and interoperability complicates the integration of AI-driven functionalities within communication systems. Furthermore, privacy and security concerns arise due to data sharing and distributed computation, necessitating robust protection mechanisms.

Motivated by the above challenges, this paper aims to provide a comprehensive overview of AI-driven 6G technology by analyzing its network architecture evolution, key enabling technologies, potential use cases, and associated challenges and opportunities. The contributions of this paper can be summarized as follows:

- This paper provides a potential evolution path for RAN architecture and proposes a new end-to-end (E2E) AI-driven network framework integrating AI capabilities as a native feature to enable efficient and scalable intelligent services.
- This study delves into critical enabling technologies, including cross-domain AI collaboration, native computing, and native security, offering practical insights for 6G implementation.
- This paper identifies the opportunities presented by AI-driven 6G networks and analyzes the challenges in achieving seamless integration, providing guidance for future research.

The remainder of this paper is structured as follows: Section 2 explores the potential evolution of RAN architecture and presents an end-to-end AI-driven network framework. Section 3 discusses key enabling technologies for AI-driven 6G networks, including cross-domain AI collaboration, native computing, and security mechanisms. Section 4 focuses on potential use cases, highlighting Network for AI scenarios such as embodied intelligence, wearable devices, and generative AI for 6G networks. Finally, Section 5 presents a discussion of future opportunities and challenges, and Section 6 presents the conclusions.

2. Evolution of AI-Driven 6G Network Architecture

Current communication systems are primarily constructed for connectivity, whereas intelligent capabilities are realized by supplementary computing resources at the edge or in the cloud. Most current research regards AI features as external to the communication system, implying that AI-driven intelligence is deployed as an overlay service. However, in the 6G era, networks need to provide both connectivity and intelligence with native AI capability. The network may evolve from a mere data transmission pipeline into a foundational platform for intelligent services. This transformation requires a fundamental redesign of the network architecture.

The evolution of RAN architecture and the management of E2E AI capabilities may become key factors in unlocking the full potential of AI within 6G networks. This chapter begins by examining the evolution of RAN architecture, exploring the various components

of RAN functionality and potential evolutionary paths. Then, this paper further proposes an E2E AI-driven network framework, discussing the requirements and solutions from the perspectives of infrastructure, network functions, digital twins, and the AI Layer.

2.1. RAN Architecture Evolution

As the network edge, the RAN is closer to end users, enabling the provision of more real-time services. Therefore, the evolution of RAN architecture upon the introduction of AI is critical to AI-driven communication systems. Both Open RAN (O-RAN) and virtualized RAN (vRAN) leverage virtualization and cloud-native technologies, adopting open and modular designs to enhance network flexibility and reduce deployment costs.

vRAN primarily focuses on using general-purpose hardware and virtualization, while O-RAN emphasizes open and standardized interfaces. O-RAN introduces near-real-time RAN Intelligent Controllers (RIC) and non-real-time RIC to enable intelligent functionalities within the RAN. However, discussions at this stage have largely centered on applying AI to optimize RAN performance, often referred to as AI for RAN.

With the establishment of the AI RAN Alliance, the scope of AI and RAN integration has expanded significantly. Beyond AI for RAN, new models include AI on RAN, which explores leveraging RAN capabilities to support edge AI applications, and AI and RAN, which focuses on the infrastructure-level integration of AI and RAN. These developments highlight the increasing role of RAN as a critical enabler in the AI-driven communication ecosystem, paving the way for more intelligent services at the network edge.

It appears that adopting general-purpose computing (GPC) platforms and cloud-based solutions is an important direction for the evolution of the RAN to natively support AI capabilities. However, compared to dedicated hardware, GPC platforms still face many challenges. As shown in Table 1, GPC platforms use transparent protocols and decouple software from hardware, offering better openness [20,21]. However, replacing dedicated baseband processing units with a general-purpose processor results in significant shortcomings in performance and energy efficiency. Dedicated hardware, on the other hand, has low energy consumption and high technological maturity, making it more suitable for current network requirements. However, it lacks openness and flexibility and is heavily dependent on specific vendors.

Table 1. Comparison between GPC and dedicated hardware.

	Dedicated Hardware	**GPC**
Performance	Highly optimized	Relatively high
Latency	Ultra-low latency	Higher latency due to virtualization and software stack limitation
Energy Efficiency	Higher—Optimized hardware consumes less power for the same workload	Lower—Higher power consumption due to software-based processing overhead
Advantages	High reliability and stability	Flexible deployment, ideal for dynamic environments, lower cost
Disadvantages	Long development cycles, higher costs, limited flexibility	Inferior performance and power efficiency, higher latency

The next-generation RAN is likely to continue adopting the architecture composed of the Centralized Unit (CU), Distributed Unit (DU), and Radio Unit (RU). Which part of the

functions needs to adopt a cloud-based architecture depends on the demand for computing resources and deployment density. Compared to in 5G networks, the deployment of DUs will become denser in 6G networks, and the demand for computing resources is not high. DUs will mainly perform tasks with high real-time requirements but low complexity. Therefore, DUs based on dedicated hardware are more cost effective. On the other hand, CUs are deployed in a more centralized manner and have higher demands for computing resources. By cloudifying CUs, it is possible to better support AI applications at the network edge and improve the flexibility of the RAN.

Since AI heavily relies on data and computing resources, the openness of computing resources and data will profoundly influence the scope and impact of AI applications in next-generation networks. Figure 2 illustrates potential options for computing resource exposure based on GPC and dedicated hardware. Dedicated hardware is typically designed for specific communication or AI functions, and its computing resources are not able to be exposed. In this case, AI capabilities can only be used to optimize the performance of the RAN itself. On the other hand, GPC-based solutions leverage virtualization technologies to flexibly construct virtualized communication and AI functions from general-purpose computing resource pools. This approach allows for the dynamic adjustment of computing resource allocation ratios according to changing workload. Additionally, idle computing resources can be exposed to third-party applications, creating new revenue models and improving resource utilization.

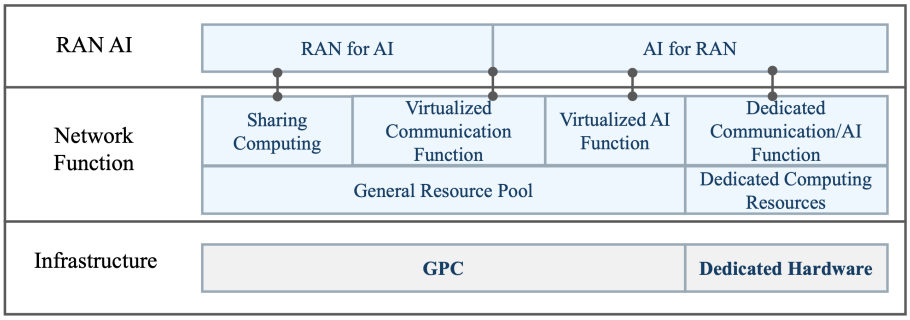


Figure 2. Possible computing resource exposure options based on GPC and dedicated hardware.

Meanwhile, data exposure requires greater caution. Currently, there are no high-quality datasets specifically for wireless networks. Most of the existing datasets were either generated through simulations or collected as long-term statistical data via network management systems. This limitation arose for several reasons: (1) Lack of standardized interfaces: The industry lacks unified standardized open interfaces, making cross-device data collection challenging. Although O-RAN has defined standardized open interfaces such as O1, O2, E2, and A1, not all vendors have adopted the O-RAN model. (2) Privacy and security concerns: Wireless data, especially user-related data, inherently involve privacy and security requirements. Without robust security mechanisms, opening such data to third parties may lead to risks such as unauthorized tampering and information leakage. (3) Ownership disputes: Data openness may involve the interests of multiple stakeholders, including equipment vendors, operators, and service providers. Disputes over data ownership and usage rights may arise. Addressing these challenges is crucial for unlocking the potential of AI in the next-generation RAN. The exposure of computing resources and data will be pivotal in determining the application scope and influence of network AI.

2.2. E2E AI-Driven Network Framework

In the 6G era, AI capabilities will not be limited to the RAN but will also be deployed across the CN, network management systems, the cloud, and even terminals. Therefore, an E2E framework designed to coordinate AI functionalities across different domains is another key requirement for AI-driven 6G technology. Figure 3 presents a reference framework for this E2E AI integration.

The network infrastructure consists of GPC platforms and dedicated hardware, including radio-frequency devices, storage devices, computing devices (e.g., Central Processing Units (CPUs), Graphics Processing Units (GPUs)), and transmission equipment. General-purpose hardware can leverage virtualization technologies to construct virtual resource pools, enabling the deployment of network functions across different domains on top of the virtualization layer. However, some network functions, such as DUs and RUs, will continue to rely on dedicated hardware to meet specific performance requirements.

The cloud–network functional domains encompass the cloud, RAN, CN, and network management systems, each embedded with AI capabilities. In the 3GPP framework, the CN introduces a new network function called the Network Data Analytics Function (NWDAF) [12] to enable data analytics. On the RAN side, AI functionalities are implemented through gNB [9], while Operations, Administration, and Maintenance (OAM) systems are responsible for AI management and utilize the Management Data Analytics Function (MDAF) for management data analysis [11]. The O-RAN introduces additional network functions, such as near-real-time RIC and non-real-time RIC, to enable AI functionalities in the RAN domain. Looking ahead to the evolution of networks, it is anticipated that, in the 6G era, components such as the CU control plane, DU, and certain CN functions will natively integrate AI capabilities. This end-to-end integration will enhance network intelligence, providing a unified platform for efficient and adaptive operation.

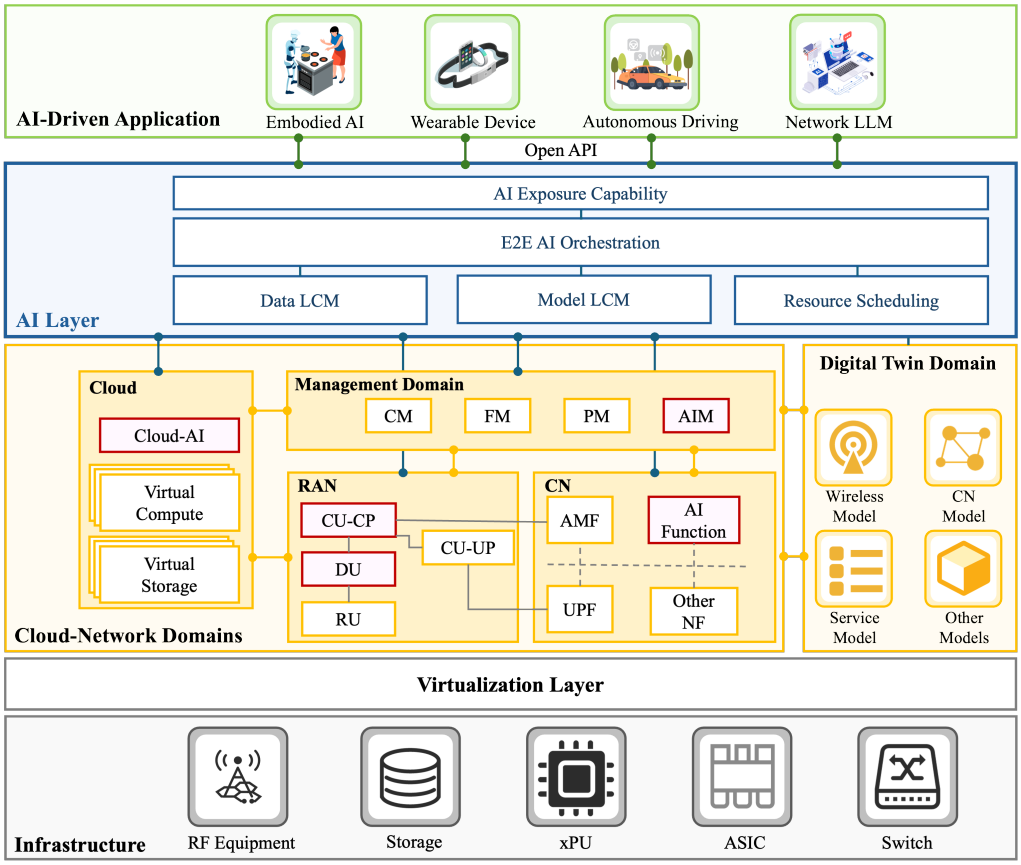


Figure 3. AI-driven E2E next-generation framework.

In addition to AI integration, digital twin technology is likely to become a critical feature of 6G networks. By enabling real-time monitoring and modeling of network environments, topologies, resources, and services, the digital twin domain provides an ideal testing environment for the execution of AI strategies and the realization of autonomous network operations. One significant challenge of applying AI in networks is its inherent lack of explainability [22]. Directly executing AI-driven strategies poses unforeseen risks to network performance and stability. By leveraging digital twin technology to pre-validate AI strategies, these potential risks can be mitigated. This capability makes digital twin technology an indispensable tool for enhancing the reliability and robustness of AI-enabled 6G networks, thereby facilitating efficient network management and operation [23].

To enable the coordination of AI capabilities across different network domains, this paper further proposes the concept of the AI Layer. The AI Layer logically connects AI functionalities distributed across the network and is responsible for AI lifecycle management and orchestration. It ensures the seamless integration of AI-related processes across the cloud, CN, RAN, and terminals, providing a unified capability for managing AI resources and tasks. The AI Layer oversees the lifecycle management of data, models, and resources. For data, it constructs flexible pipelines and datasets tailored to the requirements of training and inference tasks, covering data collection, processing, classification, storage, and usage. For models, it manages training, validation, testing, deployment, and inference, while also supporting advanced operations such as splitting and merging for large language models (LLMs). Resource scheduling is another critical function, involving the dynamic allocation of communication resources on a spectrum to balance workloads between traditional and AI-enabled services, as well as managing computing resources such as CPUs and GPUs to meet the demands of AI tasks.

Additionally, the AI Layer provides end-to-end orchestration, enabling it to match and assemble AI capabilities dynamically based on specific task requirements. Compared to the existing works of 3GPP and O-RAN, our proposed framework employs the AI Layer to integrate AI capabilities from different domains, including management, the CN, the RAN, and the terminal. In the E2E framework, the AI capabilities are natively deployed. By exposing AI capabilities through open Application Programming Interfaces (APIs), including computing resources and models, the AI Layer supports third-party applications, enhancing the network's AI service capabilities and promoting collaboration within the AI ecosystem. This approach not only maximizes the efficiency of AI resource utilization but also fosters the development of an open and interoperable AI-driven network architecture.

3. Key Enabling Technologies for AI-Driven 6G Networks

This section examines the key enabling technologies necessary for AI-driven 6G networks. We investigate cross-domain AI collaboration, native computing, and native security mechanisms, which are essential for facilitating an AI ecosystem within 6G networks.

3.1. Cross-Domain AI Collaboration

AI capabilities in 6G networks will be deployed in both distributed and centralized ways. AI on the RAN side may be distributed across the CU and DU, while, on the CN side, NWDAFs will be deployed in a distributed manner. Data centers and the cloud will also provide centralized computing and AI services, such as large language models. Therefore, cross-domain AI collaboration will be a natural requirement for AI-driven 6G. This collaboration will be reflected not only in the cooperation between networks and terminals, but also in the cooperation between physical domains and digital twin domains. Figure 4 illustrates the collaboration of AI across multiple domains, where each domain will provide AI control and management within its own domain and the network

management system will offer end-to-end AI orchestration domain management to match the requirements and computing resources of different AI tasks from a global perspective.

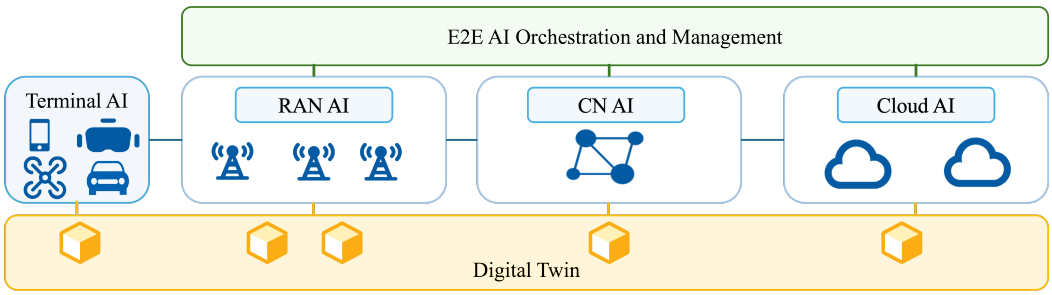


Figure 4. AI collaboration across different domains.

The distributed deployment of AI across different domains may cause numerous challenges for 6G technology, especially in terms of data security and privacy issues. Distributed learning has emerged as a critical technique for incorporating AI algorithms into 6G network systems which can process and analyze substantial data across geographically distributed environments through edge computing. In the typical distributed learning paradigm, the training and inference tasks are distributed across multiple computational nodes, which may range from edge devices to cloud servers. The advantage of distributed learning lies in its effectiveness in harnessing the computational power of multiple devices while preserving data privacy.

Furthermore, with the introduction of generative AI technologies, 6G networks have a stronger demand for collaboration between models. Large models require massive numbers of data and computing resources to support the training of large-scale parameter models, typically deployed in central nodes or the cloud. However, large models generally offer general AI capabilities, such as text interaction and semantic understanding. When addressing specific scenarios within different domains of the network, general large models cannot independently meet the AI requirements, and collaboration with domain-specific smaller models is inevitable. Therefore, the coordination between large models and small models has become a research hotspot. By deploying the large models in the cloud and the small models on the edge, a hybrid edge–cloud collaborative model inference architecture is proposed [24]. The inference results of small models, denoted as the probability of draft tokens, are compared with the results of large models, which can give a quick response on whether to accept or reject the draft tokens. If the draft token is rejected, resampling is executed by large models to generate the final output [25]. It showcases a promising and feasible solution that enables telecom operators to deploy large models for mobile devices which can take full advantage of the high inference capability of large models while leveraging the distributed computing resources on the devices.

3.2. Native Computing

As a key enabler for 6G applications, the native computing network is an integrated communication and computation technical system which can provide connection–computation–application services. By decoupling the computing power of wireless base stations and the communication services, native computing can enable seamless, intelligent, and efficient operations in 6G networks. Through the intelligent scheduling capability of native computing, base stations can support the communication services and computing services at the same time. Native computing typically involves the following two key techniques:

(1) Computing capacity sensing: As resources can be provisioned dynamically based on demand, the computing capacity-sensing technique highlights the capability to monitor

and assess the available computing resources in the system at any given time. By employing virtualization technology, the idle computing power of baseband units (BBUs) will be virtualized as the computing resource pool to support computing applications other than communication services. The virtualized resources can be measured with various metrics, for instance, CPU usage, memory availability, storage space, and network bandwidth. In this regard, native computing can give advice on how to allocate computing resources efficiently, especially in cloud computing and edge computing environments. As the edge devices are equipped with limited computing resources in the edge computing scenario, computing capacity sensing can help manage the load between local processing and remote cloud computing.

(2) Computing resource scheduling: In the data-center-based architecture, multiple servers or mobile users need access to the shared computing resources, and thus the design of computing resource scheduling strategy becomes crucial. By dynamically managing and allocating computing resources that have been sensed based on target tasks or workloads in the system, the computing resource scheduling technique aims to optimize the network performance and maximize the user's quality of experience. The key components of computing resource scheduling are tasks, workloads, resources, and schedulers, in which tasks or workloads may vary in complexity, execution time, and resource requirements. The resources include physical and virtual resources, including processors, memory, disk storage, and network bandwidth, which need to be allocated to tasks. The schedulers are software components or algorithms that manage the allocation of resources to tasks, for example, the time series prediction algorithms, which prioritize tasks, allocate resources, and track the progress of running tasks. In [26], native computing is utilized to support split AI applications, to reduce inference time, network traffic load, and user energy consumption. In [27], the impact of AI-native architectures on throughput, latency, and efficiency is studied through quantitative simulation results, which are highly correlated with our works.

3.3. Native Security

With the emergence of advanced capabilities in 6G networks, robust and comprehensive security mechanisms have become imperative. Thus, the concept of native security has been proposed and widely studied. Native security typically refers to the principle of embedding security mechanisms directly into the foundational architecture of the network, which leverages AI, blockchain, and quantum cryptography to create a resilient and trustworthy communication infrastructure [28].

By integrating AI technologies into the core network, AI-driven native security can provide autonomous, intelligent, and adaptive security solutions to the 6G system which leverage advanced AI techniques to identify anomalies, predict vulnerabilities, and respond to attacks dynamically. Specifically, by analyzing vast numbers of data, AI techniques such as anomaly detection and supervised learning can identify unusual patterns and help the system take corrective actions automatically without human intervention [29]. Additionally, the anomaly detection and blockchain-based authentication mechanisms are also key enablers for AI-driven system security [30,31].

Benefiting from AI-driven native security, diverse applications have been developed. AI-empowered intrusion detection and prevention systems can monitor network traffic and identify malicious activities such as distributed denial-of-service (DDoS) attacks and unauthorized access, with AI algorithms providing the ability to classify normal and abnormal behavior with high accuracy [32]. Moreover, AI-driven native security systems can detect and mitigate malware by analyzing code signatures and behavioral patterns, employing convolutional neural networks to identify sophisticated malware that evades traditional signature-based detection methods [33]. In addition, quantum-resistant

cryptography has been regarded as a promising solution for 6G networks. In [34], an AI-driven system is proposed to analyze quantum algorithms and their potential impact on traditional encryption which can design robust cryptographic techniques that withstand quantum threats.

However, despite the potentials illustrated, AI-driven native security still faces several challenges. AI systems are vulnerable to adversarial attacks where attackers manipulate input data to deceive the AI models and prompt false decisions. Moreover, AI models require access to large datasets for training and operation, and thus data privacy concerns escalate. The high computation and communication complexity of AI models also becomes a heavy burden, especially for the resource-constrained edge devices in 6G networks.

4. Use Cases Analysis

AI for Network scenarios, employing AI technology to enhance network performance, have been extensively researched in both industry and academia. This chapter concentrates on Network for AI solutions, examining three representative use cases, wearable devices, embodied intelligence, and generative AI, to demonstrate how AI-driven networks facilitate smarter, more secure, and efficient communication and AI services through AI collaboration across the cloud, the edge, and devices. Wearable devices address human health and entertainment needs, and embodied intelligence extends AI applications to robotic systems. Meanwhile, generative AI has the potential to introduce new capabilities and revenue models for future networks.

4.1. AI for Wearable Devices

With the rapid growth of AI technology and edge computing, network intelligence is gradually moving towards the edge, giving rise to AI-enabled wearable devices. 'Wearable devices' typically refers to electronic devices worn on the body, which are employed to monitor and transmit collected information about consumer behaviors and activities in an environment. Wearable devices often integrate various sensors, such as temperature sensors, accelerometers, gyroscopes, and heart rate sensors, to collect personal data and improve daily life. As shown in Figure 5, the base station collects data from the wearable devices, which may encompass earphones, watches, glasses, and footwear. Empowered by AI chips embedded at the base station, user terminals will be provided with real-time intelligent analysis, such as route advice on avoiding a traffic accident ahead.

In the smart wearable device use case, the advantages of the proposed AI-native RAN mainly lie in two aspects. First, traditional wearable devices are equipped with sensors and cameras to collect data, and the RAN only provides connectivity to route data to the cloud for processing. In contrast, with the AI-native RAN, AI models deployed within the RAN itself can process RAN-side data and extend the environmental perception function of wearable devices beyond what the onboard sensors can capture. Furthermore, different from the conventional edge computing technique which allocates computation tasks to the MEC, the AI-native RAN can directly process the data at the base station without UPF-based data offloading. Thus, the transmission latency can be saved and the real-time AI service efficiency can be significantly improved.

In the 6G era, smart wearables will be further accelerated by the ultra-low-latency service provided by 6G networks, which is important for some real-time applications such as medical equipment and health monitoring. Moreover, the high data speed of 6G networks can give the users immersive experiences with the help of smart wearable devices, such as smart glasses, which facilitate AR/VR applications. Additionally, as a key feature of 6G networks, holographic communication could revolutionize the way wearable devices

are used for communication, and enable real-time interaction such as immersive virtual meetings and real-time holographic displays.

Figure 5. RAN AI for wearable devices.

The recent literature on AI-driven wearable devices has been devoted to utilizing AI technology to enhance the reliability, accuracy, and efficiency of wearable devices. Convolutional neural networks have demonstrated significant advantages in human activity recognition and health monitoring, providing real-time health indicator analysis and improving recognition accuracy. Moreover, recurrent neural networks (RNNs) are employed in processing time series data and handle long-term dependencies through their memory mechanisms, making them suitable for analyzing continuous motion data collected by accelerometers and gyroscopes. Further, as the data captured by sensors may come from different sources, including vision, sound, motion, physiological signals, multimodal characteristics have shown great potentiality for developing AI algorithms. AI-driven wearable devices integrate data from multiple modalities and leverage the complementarity of multimodal data to enhance perception and interaction capabilities.

We take smart glasses as an example to illustrate the coordination of AI models in the terminal, edge, and cloud domain based on our proposed framework. The smart glasses first collect real-time data, including images, texts, audios, and posture data, and upload the data to the edge server. The user can select the AI services on the smart glasses. The edge server, typically deployed near the CU, employs tiny AI models to process the data and execute real-time tasks such as image recognition and text translation. The AI Layer serves as the service orchestrator for multiple AI services from different users. It determines whether the cloud LLM is activated for the AI service, and routes the corresponding data to the cloud. It also provides a strategy to update the models on the edge. If activated, the cloud employs an LLM for more complex tasks such as smart recommendations, network optimization, and intelligent chatbots. The cloud sends feedback to the smart glasses to be displayed to the users.

It should be noted that data privacy in the wearable device use case is essential, as the collected data may be correlated to users' health, wealth, and behaviors. Therefore, the data security mechanism should be enforced to prevent data leakage. For example, the AI Layer works on data lifecycle management, and it can determine what kind of data can be

uploaded to the cloud and what kind should be processed locally. When it detects data leakage, some effective measures should be taken to protect user privacy.

4.2. Embodied AI

Embodied intelligence, a critical component of future intelligent systems, currently relies heavily on terminal-based intelligence. This dependency results in high device costs, limited AI capabilities, and suboptimal energy efficiency, significantly constraining user experience. The integration of AI with RAN introduces a transformative paradigm, enabling a redistribution of intelligence with cloud–edge–terminal collaboration. By offloading general-purpose AI functionalities to the RAN, this approach reduces terminal complexity and cost while enhancing overall user experience and scalability.

As shown in Figure 6, terminal AI focuses on executing latency-sensitive and context-aware tasks, such as real-time sensor data processing, localized decision-making, and direct user interaction. RAN AI, on the other hand, handles computationally intensive yet generalized functions, including multi-device coordination, mobility management, and adaptive resource allocation, leveraging its proximity to end users for improved responsiveness. Cloud AI, positioned at the top of the hierarchy, performs high-level functions such as model training, large-scale data analysis, and global optimization, ensuring system-wide intelligence enhancement and sustainability. This hierarchical collaboration enhances energy efficiency at the terminal level, enables dynamic adaptability and scalability through RAN-based intelligence, and ensures continuous learning and improvement via cloud integration. Together, these innovations establish a robust framework for supporting intelligent and cost-effective embodied systems in future AI-driven networks.

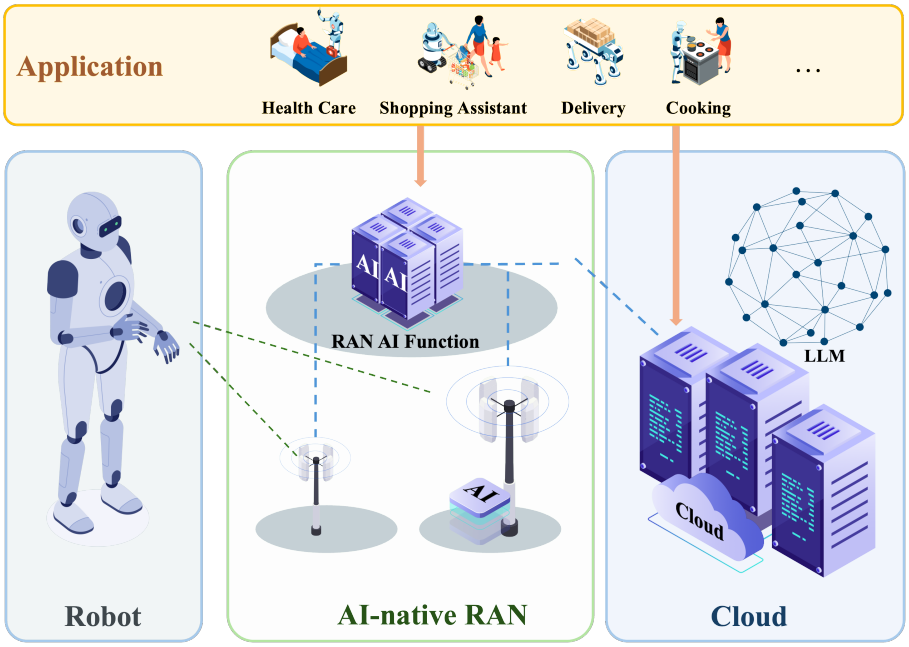
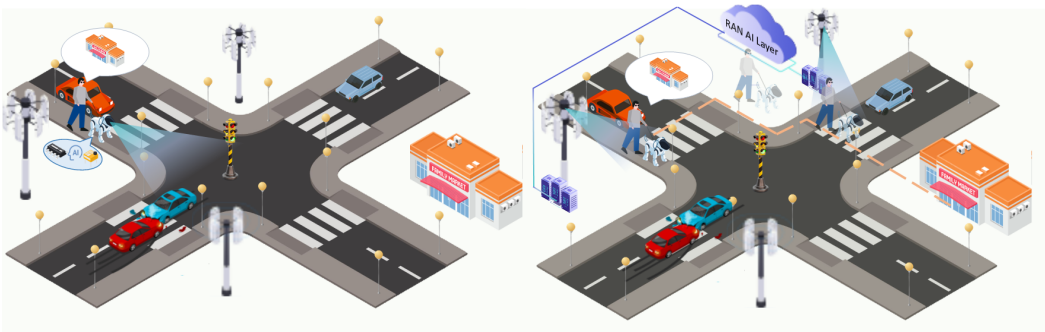


Figure 6. Cloud–edge–device collaboration for embodied intelligence.

Figure 7 illustrates an example of an AI-driven network for robotic guide dogs, demonstrating how cloud–edge–terminal AI collaborates to enhance intelligent assistance services for robotic guide dogs. Approximately 43 million individuals are blind globally; however, only about 22,000 guide dogs are actively employed to assist this population. Robotic guide dogs, enhanced by AI technologies, present a promising alternative. These robotic systems, typically powered by embedded units such as Raspberry Pi or Jetson, handle tasks like obstacle detection and path planning. However, these systems are constrained by limited

computing resources and insufficient adaptability to dynamic environments. The AI-native RAN offers a possible solution by offloading intensive AI processing tasks to distributed RAN nodes. This approach not only reduces energy consumption and computing pressure on terminals, but also addresses the social challenges of providing more accessible assistance to persons with disabilities. Without the AI-native RAN, all AI capabilities are concentrated on the terminal, requiring it to handle sensing, decision-making, path planning, and human–machine interaction independently. This imposes extremely high requirements on the device's computing capacity and performance. Compared to terminals, the RAN can perceive and acquire environmental information on a larger scale. With the assistance of RAN AI, it can execute AI tasks such as path planning, incident detection, and localization, and provide smarter and safer services for visually impaired individuals.



(**a**) (**b**)

Figure 7. Comparison: (**a**) Robotic guide dog without AI-native RAN. (**b**) Robotic guide dog with AI-native RAN.

4.3. Generative AI Within 6G Networks

With the rapid development of generative AI technologies, large language models (LLMs) and multimodal models have achieved remarkable success in natural language processing (NLP) and image and video generation. Introducing generative AI into communication systems has attracted significant attention from academia and the industry, particularly in areas such as intelligent operations and management and network automation. By leveraging the strong comprehension capabilities of LLMs, generative AI can enhance intent recognition and translation accuracy in intent-driven networks.

Compared to traditional AI models, generative AI leverages massive parameter scales and extensive data training to improve understanding and generalization capabilities. Consequently, LLMs are typically deployed in data centers or cloud environments. Many companies provide flagship models alongside lightweight versions to cater to diverse user needs. Table 2 summarizes the capabilities and revenue models of several commonly used LLMs and multimodal models. For text-focused LLMs, the predominant pricing model involves token-based billing, where output tokens are priced higher than input tokens. For multimodal models, fees are often associated with converting images or videos into tokens, such as Gemini, or charged based on the number of images and video duration, as seen with Imagen3 and Text to Speech (TTS). Some text-to-image generation models, such as Stable-Diffusion-1.5, charge based on model units and usage duration.

These business models provide valuable insights into integrating generative AI with communication systems. Deploying large models at central network nodes or lightweight models at the network edge, including the RAN, presents opportunities for the telecom industry to introduce new revenue opportunities. Beyond traditional data traffic income,

operators could monetize computational power, AI models, or tokens, creating innovative ways to enhance network profitability.

Table 2. Analysis of revenue models for different LLMs and multimodal models.

Models	Type	Capabilities (Tokens)		Pricing (Dollars/Ktokens)		
		Max Input	Max Output	Rules	Input	Output
GPT-4o [35]	Text	128,000	16,384	charge by tokens	0.0025	0.01
Claude3.5 Sonnet [36]	Text	200,000	8192	charge by tokens	0.003	0.015
Llama-3 [37]	Text	6500	1500	charge by tokens	0.00065	0.00275
qwen-max [38]	Text	32,768	8192	charge by tokens	0.0027	0.0082
hunyuan-large [39]	Text	28,000	4000	charge by tokens	0.00055	0.0016
Doubao-pro [40]	Text	256,000	4000	charge by tokens	0.00068	0.0012
Gemini 1.5 Pro [41]	Image	-	-	charge by image	0.000329	-
	Video	-	-	charge by minute	0.019725	-
	Audio	-	-	charge by minute	0.001875	-
DALLE 3 [42]	Image	-	-	charge by image	-	0.04
Imagen 3 [43]	Image	-	-	charge by image	-	0.04
Whisper [44,45]	Speech to Text	-	-	charge by minute	0.006	-
TTS [45]	Text to Speech	-	-	charge by tokens	0.06	-
DeepSeek-R1 [46]	Text	64,000	8000	charge by tokens	0.0006	0.0022

Generative AI deployed within networks will inevitably fall short of meeting all network intelligence requirements. Therefore, the foundational intelligence provided by large models must collaborate with various small or specialized models to address the diverse intelligent tasks across different network domains [47,48]. Generative AI can act as a selector to match models with tasks based on the specific requirements of intelligent tasks and the capabilities of models deployed within the network. By understanding the task's complexity and resource constraints, large models can allocate appropriate AI models and computational resources to ensure efficient task execution. Conversely, generative AI can also serve as a global decision-maker for small models. In this role, it aggregates the outputs from specialized or small models deployed across different domains and integrates them to provide comprehensive analysis or overarching decisions. This two-way collaboration ensures that generative AI and specialized models work together seamlessly, supporting the diverse and complex scenarios required by intelligent 6G networks.

5. Opportunities and Challenges

This section explores both the promising opportunities and the critical challenges associated with the integration of AI in 6G networks. We discuss the new revenue models that AI-driven 6G networks could bring to operators and their potential to foster ecosystem development, as well as the challenges this integration poses in terms of cross-domain and cross-vendor collaboration and AI management.

5.1. Opportunities

(1) Creating a new revenue model: The integration of AICT empowers operators to explore new commercial opportunities through its AI capabilities. In the 6G era, AI as a Service (AIaaS) refers to the delivery of AI capabilities as a network-integrated service,

which enables users and enterprises to access AI models, computing resources, and AI-driven functionalities without the need for dedicated local AI infrastructure. Operators will be able to expand beyond traditional communication services by offering intelligent services such as real-time health monitoring, AI-driven virtual assistants, autonomous vehicles, and immersive AR/VR experiences. Furthermore, operators can generate new revenue streams by selling tokens or leasing underutilized network computing resources. Finally, based on the dynamic demands of communication and intelligent services, operators will be able to adjust the allocation of network computing resources in real time, optimizing resource utilization and reducing both operational and infrastructure costs.

(2) Foster the growth of the AICT ecosystem: The integration of AICT could transform networks from traditional communication service pipelines into intelligent service platforms, fostering collaboration and growth within the telecom ecosystem. Operators, vendors, service providers, software developers, and cloud providers can all benefit from this transformation. Once the network possesses AI capabilities, computational tasks at the terminal level can be offloaded to the network edge, reducing terminal costs and extending battery life. The network itself will become the foundational platform for AI services, offering models and computing resources to third parties. Additionally, this integration will strengthen the computational capability at the network edge, potentially leading to a boom in the application market, much like the emergence of smartphones did for mobile apps.

In order to foster the AICT ecosystems, operators are likely to play a key role. As the actual network operators, they serve as the bridge between vendors, service providers, and users. Operators need to integrate resources and explore new collaboration models to jointly develop technology routes, standards, and interface protocols to ensure the construction of an independent technology system. Moreover, it is critical to establish a standard tracking mechanism to promote multi-party cooperation and the standardization of key technologies such as compatibility interfaces and open APIs.

5.2. Challenges

(1) Cross-domain and cross-vendor interoperability: The high complexity of the AI-driven 6G framework depends on seamless coordination among multiple vendors across various network domains and IT areas, requiring standardized and open APIs. However, the gap between the networking and AI industries makes standardization difficult. Moreover, the integration of AI models across multiple domains (such as the core, RAN, and terminal) from various vendors complicates data and model coordination. These interoperability issues can lead to concerns around data privacy and security, as sensitive information may be exchanged between domains with different security protocols, increasing the risk of unauthorized access.

(2) E2E AI orchestration and management: The AI capabilities of a network are highly complex, with differing characteristics across data, models, and computing resources in various domains, making unified management extremely challenging. Furthermore, exposing network or cloud capabilities to third parties requires great caution and a well-defined architecture to ensure security and accountability. Additionally, the orchestration of AI components must be integrated with both cloud and network management systems. The key issue here which requires further research and discussion is whether a single management system should oversee both network and cloud-based AI or whether separate orchestration and management systems are needed.

(3) AI trustworthiness and explainability: In the 6G system, the AI-driven solutions need to be transparent and secure to avoid security risks in resource allocation and network optimization. Moreover, explainable AI techniques are crucial for making AI decisions more interpretable and thus enhance the reliability of 6G networks.

6. Conclusions

The integration of AI into 6G networks represents a paradigm shift, transforming communication systems from traditional data transmission platforms into intelligent, autonomous service infrastructures. The concept of AICT, which integrates AI, IT, and CT, was introduced to enable seamless cloud–edge–terminal collaboration. This paper analyzes the potential evolution path of RAN architecture and then proposes an E2E AI-driven network framework. Additionally, the paper examines key technologies, such as cross-domain AI collaboration, native computing, and AI-driven security mechanisms, highlighting the challenges and opportunities in developing an AI-native 6G ecosystem. Looking forward, AI-driven 6G networks will enable a wide range of intelligent applications, such as embodied intelligence, AI-enhanced wearable devices, and generative-AI-assisted services. The hybrid centralized–distributed AI deployment model will reshape network operations, service models, and business strategies, paving the way for a scalable, secure, and intelligent AI-native 6G ecosystem.

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